Natural-79 equation: source-only candidate

Fourteen nonnegative integer unknowns; 79 binary-operation abbreviations.

The unknowns, all distinct, are $J, F, A, f, i, T, w, s, r, e, z, y, g, h \in \mathbb{N}$. All monospaced names below are polynomial abbreviations, and β, K, μ, ω are fixed integers defined later.

$\texttt{valpha} = A + 1$
$\texttt{vi} = i + 1$
$\texttt{vT} = T + 1$
$\texttt{vs} = s + 1$
$\texttt{veta} = e + 1$
$\texttt{vqm} = J \cdot \beta$
$\texttt{vq} = 1 + \texttt{vqm}$
$\texttt{vX} = \texttt{vq} \cdot w$
$\texttt{vq2} = \texttt{vq} \cdot \texttt{vq}$
$\texttt{vq3} = \texttt{vq} \cdot \texttt{vq2}$
$\texttt{vY} = \texttt{vq3} \cdot \texttt{vs}$
$\texttt{vU} = \texttt{vY} \cdot \texttt{vq2}$
$\texttt{vqF} = \texttt{vqm} - F$
$\texttt{vC} = \texttt{vqF} - \texttt{valpha}$
$\texttt{vE} = \texttt{vX} \cdot \texttt{vY}$
$\texttt{vkbase} = \texttt{veta} + z$
$\texttt{vk} = \texttt{vkbase} + 1$
$\texttt{vkY} = \texttt{vY} \cdot \texttt{vk}$
$\texttt{vc} = \texttt{veta} + \texttt{vkY}$
$\texttt{vc2} = \texttt{vc} \cdot \texttt{vc}$
$\texttt{va} = \texttt{vE} + \texttt{vY}$
$\texttt{vg15} = 4 \cdot \texttt{va}$
$\texttt{vH} = 3 + \texttt{vg15}$
$\texttt{vg17} = \texttt{va} \cdot \texttt{va}$
$\texttt{vDelta} = \texttt{vH} + \texttt{vg17}$
$\texttt{vg19} = \texttt{va} \cdot \texttt{vc}$
$\texttt{vb0} = \texttt{vX} + \texttt{vg19}$
$\texttt{vell} = \texttt{vE} \cdot \texttt{vkY}$
$\texttt{vg26} = \texttt{vX} + K$
$\texttt{vg27} = \texttt{vC} \cdot \texttt{vg26}$
$\texttt{vguard\_base} = \texttt{vg27} - F$
$\texttt{vg30} = \texttt{vell} + \texttt{vk}$
$\texttt{vg38} = r \cdot r$
$\texttt{vg40} = \texttt{vell} \cdot \texttt{vg30}$
$\texttt{vg44} = \texttt{vqF} \cdot \texttt{vqm}$
$\texttt{vG} = \texttt{valpha} + \texttt{vg44}$
$\texttt{vg46} = \texttt{vq2} - 1$
$\texttt{vg47} = \texttt{vG} \cdot \texttt{vg46}$
$\texttt{vg48} = \omega \cdot \texttt{vq}$
$\texttt{vg49} = \mu + \texttt{vg48}$
$\texttt{vg50} = J \cdot \texttt{vg49}$
$\texttt{vR} = \texttt{vg47} + \texttt{vg50}$
$\texttt{vr0} = \texttt{vkbase} - \texttt{vR}$
$\texttt{vg54} = \texttt{vU} \cdot g$
$\texttt{vshift} = \texttt{vg54} - \texttt{vr0}$
$\texttt{vg58} = \texttt{vU} \cdot \texttt{vb0}$
$\texttt{vg59} = \texttt{vH} \cdot \texttt{vshift}$
$\texttt{vDstar} = \texttt{vg58} + \texttt{vg59}$
$\texttt{vg61} = \texttt{vDstar} \cdot \texttt{vDstar}$
$\texttt{vg62} = \texttt{vU} \cdot \texttt{vU}$
$\texttt{vg63} = \texttt{vDelta} \cdot \texttt{vc2}$
$\texttt{vg64} = 1 + \texttt{vg63}$
$\texttt{vg65} = \texttt{vg62} \cdot \texttt{vg64}$
$\texttt{vA1} = \texttt{vg61} - \texttt{vg65}$
$\texttt{vf2} = f \cdot f$
$\texttt{vrho} = \texttt{vf2} - 1$
$\texttt{vQ} = \texttt{vDelta} \cdot \texttt{vrho}$
$\texttt{vI} = \texttt{vc2} \cdot \texttt{vi}$
$\texttt{vg71} = \texttt{vI} \cdot \texttt{vI}$
$\texttt{vg72} = \texttt{vg71} - \texttt{vQ}$
$\texttt{vg74} = \texttt{vT} \cdot \texttt{vc}$
$\texttt{vg75} = \texttt{vg74} - \texttt{vR}$
$\texttt{vg76} = \texttt{vf2} \cdot \texttt{vg75}$
$\texttt{vV} = \texttt{vg76} - \texttt{vc}$
$\texttt{vy2} = y \cdot y$
$\texttt{vV2} = \texttt{vV} \cdot \texttt{vV}$
$\texttt{vg81} = \texttt{vy2} - \texttt{vV2}$
$\texttt{vg82} = \texttt{vQ} \cdot \texttt{vg81}$
$\texttt{vFa} = \texttt{vy2} - \texttt{vg82}$
$\texttt{vbt\_quotient} = \texttt{vqm} \cdot h$
$\texttt{vguard} = \texttt{vguard\_base} - \texttt{vbt\_quotient}$
$\texttt{vF0} = \texttt{vg38} - \texttt{vg40}$
$\texttt{vF1} = 1 + \texttt{vA1}$
$\texttt{vFs} = 1 + \texttt{vg72}$
$\texttt{vP01} = \texttt{vF0} \cdot \texttt{vF1}$
$\texttt{vP012} = \texttt{vP01} \cdot \texttt{vFa}$
$\texttt{vP0123} = \texttt{vP012} \cdot \texttt{vFs}$
$\texttt{vP01234} = \texttt{vP0123} \cdot \texttt{vguard}$
$\texttt{vout} = \texttt{vP01234} - 1$

The single equation is $\boxed{\texttt{vout} = 0}$.

The four fixed integers. All numerals, including the complete seed below, use ordinary base-10 notation. Put $\rho = 2^{67}$, $\beta = \rho^{8807} - 1$, and let $d_j = \lfloor P/10^{16j} \rfloor \bmod 10^{16}$ for $0 \le j < 943$. Let the ordered list $(\lambda_0, \dots, \lambda_{40})$ be

$$1023, 1024, 1025, 1026, 1027, 1028, 1029, 1030, 1013, 1014, 1015, 1016, 1017, 1018,$$
$$1019, 1020, 1021, 1003, 1004, 1005, 1006, 1007, 1008, 1009, 1010, 983, 984, 985,$$
$$986, 987, 988, 989, 990, 991, 1031, 1022, 1011, 992, 1095, 2195, 3296$$

Define

$$\begin{aligned} K = {} & \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ & + \sum_{j=0}^{942} d_j (\rho^{4393-j} + \rho^{5495-j}) \\ & + \sum_{j=0}^{40} 2^{10+j} (\rho^{4393-\lambda_j} + \rho^{5495-\lambda_j}). \end{aligned}$$

For integer intervals $[a, b] = \{a, a+1, \dots, b\}$, let

$$\begin{aligned} S = {} & ([0, 942] \cup [983, 992] \cup [1003, 1011] \cup [1013, 1031] \\ & \cup \{1063, 1064, 1095, 2195, 3296\}) \setminus \{1\}. \end{aligned}$$

Put $\mu = \beta - \sum_{k \in S} \rho^k - 2\rho^{1063}$ and $\omega = \beta + H$, where

$$\begin{aligned} H = 4 + \sum_{k=1013}^{1022} \rho^k + \sum_{k=1065}^{1073} \rho^k + \rho^{2195} + \rho^{3296} \\ + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) + (2^{65} - 1) \sum_{k=8792}^{8804} \rho^k + (2^{19} - 1)\rho^{8805}. \end{aligned}$$

Complete decimal integer P. Concatenate these lines in order, without spaces or line breaks:

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This is an unelaborated source-mapping draft; it is not a claim that the equation or its iPRA equivalence has been checked by Lean.